POTASSIUM CHLORIDE

High-alert — never IV bolus

Concentrate restricted

CHEMICAL NAME KCl (1 g ≈ 13.4 mmol K^+); premixed IV bags (e.g. 20 or 40 mmol/L) and concentrate (2 mmol/mL)

CLASS Electrolyte replacement (potassium salt)

TRADE NAMES Sando-K (effervescent) · Slow-K · Kay-Cee-L (liquid) · KCl premixed infusion bags

THERAPEUTIC USES / INDICATIONS

- Hypokalaemia — treatment and prevention (diuretics, vomiting, diarrhoea, DKA, refeeding).
- Maintenance K^+ in IV fluids.
- Digoxin toxicity with hypokalaemia (with caution).

ADMINISTRATION

- **PO** effervescent tablets dissolved in water, or liquid, with or after food; MR tablets swallowed whole upright (GI ulceration).
- **IV only diluted, via infusion pump** — use premixed bags; never IV push/bolus (cardiac arrest).
- Rate usually ≤10 mmol/h; faster rates only with ECG monitoring; concentrated solutions into a large or central vein.
- Concentrated KCl ampoules stored in controlled locked area.

DOSAGES

- **Mild hypokalaemia (oral route possible):** PO 40–100 mmol/day in 2–5 divided doses (max 40 mmol per dose).
- **Severe or unable to take PO:** IV, usually ≤10 mmol/h (e.g. 20 mmol over 2–3 h with ECG); max 15–20 mmol/h; premixed 0.3% KCl = 40 mmol/L.
- **Maintenance:** ~1 mmol/kg/day (adults typically 20 mmol/day orally).
- Correct hypomagnesaemia too (resistant hypokalaemia).

PHARMACOKINETICS

- Well absorbed orally.
- Moves into cells (insulin, β-agonists, alkalosis) — serum level reflects shifts.
- Excreted by kidneys (90%).

NURSING MANAGEMENT

- Two-person check of IV potassium (concentration, volume, rate, pump).
- Check K^+ before starting and repeat within 4–6 h (or per protocol) during IV replacement; check Mg^{2+} and renal function.
- Cardiac monitoring for rates >10 mmol/h or severe hypokalaemia.
- Check IV site hourly; urine output ≥0.5 mL/kg/h.
- Never add KCl to a hanging bag; mix thoroughly.

PHARMACODYNAMICS

- Restores **intracellular and extracellular potassium** → normal membrane potential in heart, nerves, muscle.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Nausea, vomiting, abdominal pain (PO).
- Pain/phlebitis at peripheral IV site.
- Diarrhoea.

ADVERSE EFFECTS (SERIOUS)

- Hyperkalaemia → peaked T waves, arrhythmias, cardiac arrest (rapid IV, renal impairment).
- GI ulceration, stricture (MR tablets).
- Extravasation tissue injury.

CONTRAINDICATIONS & CAUTIONS

- Hyperkalaemia; severe renal impairment/anuria (without monitoring).
- Untreated Addison's disease; severe dehydration.
- MR tablets in oesophageal/GI obstruction.

KEY INTERACTIONS

- **ACE inhibitors, ARBs, spironolactone, amiloride, trimethoprim, NSAIDs, ciclosporin, heparin:** hyperkalaemia.
- **Anticholinergics:** ↑ GI injury (solid forms).

PATIENT EDUCATION

- Take with food and a full glass of water; dissolve effervescent tablets fully.
- Report palpitations, muscle weakness, black stools or difficulty swallowing.
- Do not use salt substitutes containing potassium unless advised.

KEY POINTS

- Potassium replacement; high-alert.
- Recheck K^+ within hours; correct Mg^{2+}.
- Never IV bolus; peripheral max 10 mmol/h.
- Hyperkalaemia → cardiac arrest.

SOURCES CHECKED **[964]** Potassium Chloride 0.15 % w/v and Glucose 10 % w/v Solution for Infusion BP (UK SmPC) **[965]** Potassium Chloride 0.3% w/v & Sodium Chloride 0.9% w/v Solution for Infusion - BP (UK SmPC) **[588]** Umpierrez et al., Diabetologia 2024 **[587]** JBDS-IP DKA guideline (2023) **[954]** NICE CG174

Full references p. 432–457